WhiteboardOS — AI-Powered Sketch-to-Code Platform

Multimodal AI Living Prototype Studio: Hand-Drawn Wireframes & Indic Voice Blueprints -> Production Web Code

THE GREAT INDIAN INTERNSHIP CHALLENGE BY GNANI.AI | TRACK: MULTIMODAL AI | Production Submission v2.0

DEVELOPER & ARCHITECT
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FOUNDATION AI ENGINES
Gnani Prisma STT & Gemini Vision

GITHUB REPOSITORY
github.com/jagadeeshchinta/Sketch2Code

1. THE CORE PROBLEM (Design-to-Code Friction)

• 3 to 5 Days Bottleneck: Converting rough whiteboard diagrams and paper wireframes into working interactive HTML/CSS prototypes requires hours of manual frontend boilerplate coding.
• Intent & Hierarchy Loss: Spatial relationships, handwritten labels, and component layout hierarchies are frequently misinterpreted during designer-to-developer handoffs.
• Vernacular Barrier: Existing developer AI tools are English-only and text-prompt driven, creating friction for vernacular Indian builders who think and ideate in Hindi, Hinglish, and Indic languages.

2. THE PROPOSED SOLUTION (WhiteboardOS Studio)

• Autonomous Generative UI Studio: Merges Gnani AI Indic Voice with Google Gemini Multimodal Vision to compile physical wireframes into production web code in under 10 seconds.
• Indic Voice-to-UI Dictation: Speak UI blueprints naturally in English or Hinglish ("Create a Kirana store with 10-minute delivery and cart in ¹") with instant spoken confirmation.
• Production Living Prototypes: Emits semantic HTML5, modular React JSX, and Liquid Glass Tailwind CSS inside an isolated sandboxed iframe with hot-reloading conversational edits.

3. PLATFORM WORKFLOW — END-TO-END MULTIMODAL PIPELINE

STAGE 1
Multimodal Ingestion
Gnani STT / WebRTC
• Real-time mic audio via Gnani Prisma v3.0 (English, Hindi, Hinglish).
• WebRTC live optical webcam scanner, paper photo dropzone, digital canvas.
• Gallery Quick Starts (Art, Kirana, Bistro, SaaS)

STAGE 2
Spatial Vision AI
Gemini Vision / OCR
• Zero-shot spatial coordinate mapping extracting 30+ UI component boundaries.
• Domain-aware OCR transcribing labels, prices, and menus (96.2% [illegible] accuracy).
• In-browser spatial contour heuristic

STAGE 3
Streaming Synthesis
SSE / Token Stream
• Server-Sent Events (SSE) stream code token-by-token into live macOS terminal.
• Liquid Glass design tokens (obsidian theme, specular ridges, [illegible] filters).
• AST self-healing guardrails whitelist

STAGE 4
Living Sandbox & TTS
Gnani Timbre TTS
• Sandboxed iframe renders clickable buttons, live search, and dynamic states.
• Gnani Timbre TTS synthesizes spoken audio feedback via natural Indian voice ("Nalini").
• Conversational Hinglish edits & 1-

4. DUAL AI ENGINE ARCHITECTURE: GNANI AI & GEMINI AI INTEGRATION

GNANI AI — ACOUSTIC & INDIC VOICE INTELLIGENCE

• **Prisma STT v3.0 (api.vachana.ai/stt/v3):** Ingests real microphone audio with 66r auto, hi-IN, en-IN).
• **Timbre TTS v2.5 (api.vachana.ai/api/v1/tts/inference):** Synthesizes 24kHz linea confirmation after prototype generation.
• **Resilient Dual Redundancy:** Auto-fallover to browser Web Speech API guarante

GEMINI AI — SPATIAL VISION & GENERATIVE UI ENGINE

• **Zero-Shot Spatial Coordinate Mapping:** Identifies 30+ UI component boundaries (navbars, cards, inputs, buttons, tables) from messy paper sketches and whiteboard drawings.
• **Contextual Code Generation:** Emits strictly typed, production-ready semantic HTML5, custom CSS, and React JSX using responseMimeType: "application/json".
• **Multi-Model Cascading Gateway:** Automated multi-tier model rotation prevents HTTP 429 quota exhaustion; backed by in-browser Canvas spatial contour fallback.

5. KEY PLATFORM CAPABILITIES

' **Voice-to-UI Studio:** Speak prompts in English/Hinglish with live waveform visualizer.
' **Spoken AI Audio Feedback:** Gnani Timbre TTS voice confirms prototype state (with navbar mute t
' **4 Indian Startup Presets:** UPI BharatPay, ApnaKirana Supermarket, Indian Bistro & BharatCloud S
' **Conversational UI Editing:** Say "Theme dark kardo aur pricing card add karo" to hot-reload prototy
' **macOS Code Terminal:** Real-time token streaming with live cost, latency, and memory telemetry H
' **Time-Travel & ZIP Export:** Immutable version history rollback & 1-click downloadable production ZI

6. TECH STACK & SECURITY

Frontend Framework: Next.js 15 (App Router, Server Actions, SSE)
Programming Language: TypeScript 5 (Strict Mode, 100% type-safe)
Indic Voice Engines: Gnani Prisma STT v3.0 & Timbre TTS v2.5
Vision Foundation: Google Gemini Vision (@google/generative-ai)
Styling & Visuals: TailwindCSS, Liquid Glass Tokens, Lucide Icons
3D & WebGL Shaders: Three.js, React Three Fiber, Silk & Molten Metal
Security Standard: 100% Server-Side API Key containment (no client leaks)

7. REAL-WORLD IMPACT, USE CASES & EVALUATION BENCHMARKS

48x Faster Prototype Velocity
Compresses 3–5 days of manual Figma-to-code sprints into a 10-second multimodal compile. Ideal for hackathons, early-stage MVPs, and rapid design validation.

Bharat Tech Democratization
Empowers 5M+ non-native English Indian founders, students, and tier-2/3 innovators to build working software by speaking in mother tongues and Hinglish.

Proven Telemetry (/eval suite)
• 96.2% component detection accuracy across 13 test cases.
• 66ms Gnani STT latency | $0.00018 token cost per run.

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